Benchmark Results

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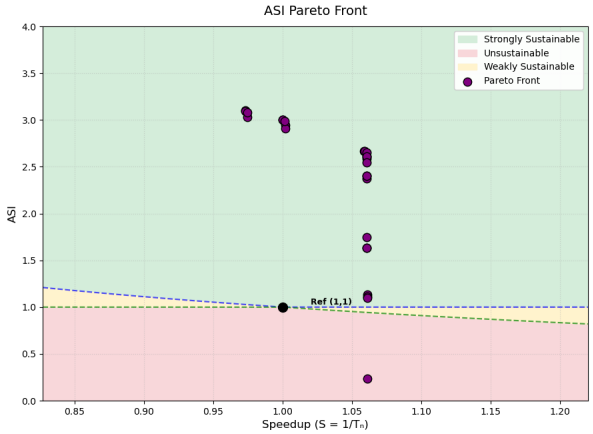
1 ML2 benchmark

ML2: L2 resident (depending on LFSR settings) linked list traversal

1.1 Greedy



Result:
Total sniper runs: 781
Final hypervolume: 3.2594

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53		4	64				512		2048	4			1024	0.2323	1.0610	67.94	141.87	Unsustainable
pentium_m		4	64	8	16		512		1024		8	2	1024	1.1008	1.0610	49.33	34.59	Strongly Sust.
pentium_m		4	64	8			512	4	1024		2	2	1024	1.1154	1.0609	47.61	34.56	Strongly Sust.
pentium_m		4	64				512		1024			2	1024	1.1342	1.0608	47.96	33.90	Strongly Sust.
a53			64				512		2048		2			1.6355	1.0608	55.79	22.20	Strongly Sust.
a53	512		64				512		2048		2			1.6355	1.0608	55.79	22.20	Strongly Sust.
a53			64				512		2048				16	1.7450	1.0607	55.19	20.90	Strongly Sust.
a53	512		64				512		1024			2		2.3749	1.0607	47.80	16.21	Strongly Sust.
pentium_m		4	64	8			512		1024			2	64	2.4038	1.0607	45.04	16.33	Strongly Sust.
a53	512	4	64	8			512		1024			2	64	2.4038	1.0606	45.04	16.33	Strongly Sust.
a53	512	4	64				512		1024			2	64	2.5447	1.0606	43.32	15.61	Strongly Sust.
a53		4	64				512	8	1024			2	64	2.5897	1.0606	41.24	15.56	Strongly Sust.
a53	512	4	64				512		1024			2	16	2.6083	1.0606	43.17	15.25	Strongly Sust.
a53		4	64				512		1024			2	16	2.6083	1.0605	43.17	15.25	Strongly Sust.
a53		4	64				512	8	1024			2	16	2.6546	1.0605	41.09	15.19	Strongly Sust.
a53		4					512	8	1024			2	16	2.6666	1.0589	40.92	15.14	Strongly Sust.
a53		4					512	4	1024			2	16	2.6682	1.0589	40.92	15.13	Strongly Sust.
a53		4	64			4			1024			2	32	2.9076	1.0019	36.49	14.31	Strongly Sust.
a53		4	64			4			1024			2	16	2.9477	1.0018	36.40	14.12	Strongly Sust.
a53		4	64					8	1024			2	16	2.9861	1.0013	34.86	14.08	Strongly Sust.
a53		4						8	1024			2	16	3.0002	0.9999	34.69	14.03	Strongly Sust.
a53		4						4	1024			2	16	3.0022	0.9999	34.69	14.02	Strongly Sust.
a53		4	64			4	128		1024			2	16	3.0325	0.9746	35.43	13.81	Strongly Sust.
a53		4	64				128	8	1024			2	16	3.0823	0.9744	33.37	13.77	Strongly Sust.
a53		4					128	8	1024			2	16	3.0970	0.9732	33.20	13.72	Strongly Sust.
a53		4					128	4	1024			2	16	3.0991	0.9731	33.20	13.71	Strongly Sust.

2 Benchmarks

2.1 ML2

ML2: L2 resident (depending on LFSR settings) linked list traversal.

Listing 1: ML2 benchmark source

```
#include <stdio.h>
#include <stdlib.h>     /* malloc, free, rand */

#include "common.h"

#define ASIZE 65536*4
#define STEP    128
#define ITERS  4096
#define LEN    2048


typedef struct dude {
  int p1,p2,p3,p4;
} dude;


dude arr[ASIZE];
__attribute__ ((noinline))
int loop(int zero) {
  int t = 0, count=0;

  unsigned lfsr = 0xACE1u;
  do
  {
      /* taps: 18 17 16 13; feedback polynomial: x^18 + x^17 + x^16 + x^13 + 1 */
      lfsr = (lfsr >> 1) ^ (-(lfsr & 1u) & 0x39000u);
      lfsr = lfsr + arr[lfsr].p1;
  //} while(++count < ITERS);
  } while(lfsr != 0xACE1u);

  return t;
}


int main(int argc, char* argv[]) {
   argc&=10000;
   ROI_BEGIN();
   int t=loop(argc);
   ROI_END();
   volatile int a = t;
}
```

2.2 CCl

CCl: Impossible control with large basic blocks (potentially larger penalty)

Listing 2: CCl benchmark source

```
    #include <stdio.h>
#include "randArr.h"
#include "common.h"

#define ASIZE 2048
#define STEP   256
#define ITERS    8

__attribute__ ((noinline))
```

```
10 int loop(int zero) {
11   int t = 0,i,iter;
12   for(iter=0; iter < ITERS; ++iter) {
13     for(i=0; i < STEP + zero; i+=1) {
14       if(randArr[i])  {
15         t+=3+3*t;
16         t&=0xFFF;
17         t+=3+3*t;
18         t&=0xFFF;
19         t+=3+3*t;
20         t&=0xFFF;
21         t+=3+3*t;
22         t&=0xFFF;
23         t+=3+3*t;
24         t&=0xFFF;
25         t+=3+3*t;
26         t&=0xFFF;
27         t+=3+3*t;
28         t&=0xFFF;
29         t+=3+3*t;
30         t&=0xFFF;
31         t+=3+3*t;
32         t&=0xFFF;
33       } else {
34         t&=0xFFF;
35         t+=3+3*t;
36         t&=0xFFF;
37         t+=3+3*t;
38         t&=0xFFF;
39         t+=3+3*t;
40         t&=0xFFF;
41         t+=3+3*t;
42         t&=0xFFF;
43         t+=3+3*t;
44         t&=0xFFF;
45         t+=3+3*t;
46         t&=0xFFF;
47         t+=3+3*t;
48         t&=0xFFF;
49         t+=3+3*t;
50         t&=0xFFF;
51         t+=3+3*t;
52       }
53     }
54   }
55   return t;
56 }
57
58 int main(int argc, char* argv[]) {
59    argc&=10000;
60    ROI_BEGIN();
61    int t=loop(argc);
62    ROI_END();
63    volatile int a = t;
64 }
```

2.3 MIP

MIP: Large Instruction Region – Instruction cache Misses

```
    #include <stdio.h>
#include "common.h"

#define ASIZE 2048
#define STEP   128
#define ITERS    4

__attribute__ ((noinline))
int loop(int zero) {
  int t = zero,i,iter;
  for(iter=0; iter < ITERS; ++iter) {
    t += 1;
    t *= 2;
    t -= 1;
    t &= 0xFFF00FFF;
    t ^= 0x14022;
    t %= 0xFFFFFF3;
    t += 6;
    t *= 10;
    t += 1;
    t *= 2;
    t -= 1;
    t &= 0xFFF00FFF;
    t ^= 0x14022;
    t %= 0xFFFFFF3;
    t += 6;
    t *= 10;
    t += 1;
    t *= 2;
    t -= 1;
    t &= 0xFFF00FFF;
    t ^= 0x14022;
    t %= 0xFFFFFF3;
    ...
    t += 1;
    t *= 2;
    t -= 1;
    t &= 0xFFF00FFF;
    t ^= 0x14022;
    t %= 0xFFFFFF3;
    t += 6;
    t *= 10;
  }
  return t;
}

int main(int argc, char* argv[]) {
   argc&=10000;
   ROI_BEGIN();
   int t=loop(argc);
   ROI_END();
   volatile int a = t;
}
```

2.4 EI

EI:Integer Execution – 8 Independent computations per iteration

```
    #include <stdio.h>
#include "common.h"

#define ASIZE 2048
#define STEP   128
#define ITERS 4096

__attribute__ ((noinline))
int loop(int zero) {
  int t1 = 1;
  int t2 = 89;
  int t3 = 3;
  int t4 = 21;
  int t5 = 2;
  int t6 = 7;
  int t7 = 2;
  int t8 = 3;
  int t9 = 1;
  int t10 = 89;
  int t11 = 3;
  int t12 = 21;
  int t13 = 2;
  int t14 = 7;
  int t15 = 2;
  int t16 = 3;

  int i;

  for(i=zero; i < ITERS; i+=1) {
    t1+=i;
    t2+=i;
    t3+=i;
    t4+=i;
    t5+=i;
    t6+=i;
    t7+=i;
    t8+=i;
    t9+=i;
    t10+=i;
    t11+=i;
    t12+=i;
    t13+=i;
    t14+=i;
    t15+=i;
    t16+=i;
  }
  return t1+t2+t3+t4+t5+t6+t7+t8*t9+t10+t11+t12+t13+t14+t15+t16;
}

int main(int argc, char* argv[]) {
   argc&=10000;
   ROI_BEGIN();
   int t=loop(argc);
   ROI_END();
   volatile int a = t;
```

```
56 }
```